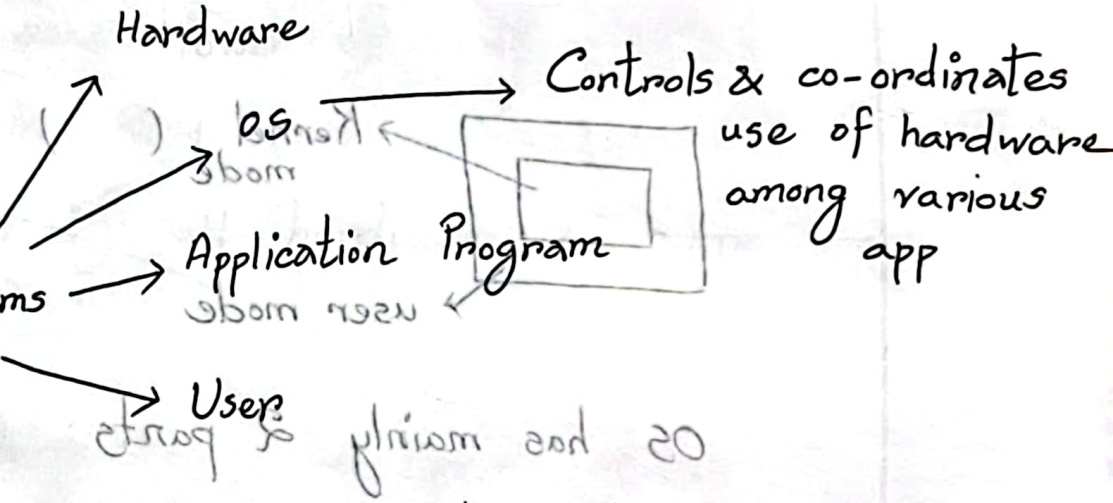
lid Syllabus

- Introduction
- Process
- Process Scheduling
- Disk Management

Book:
OS
↳ Silberchatz

Chap-01

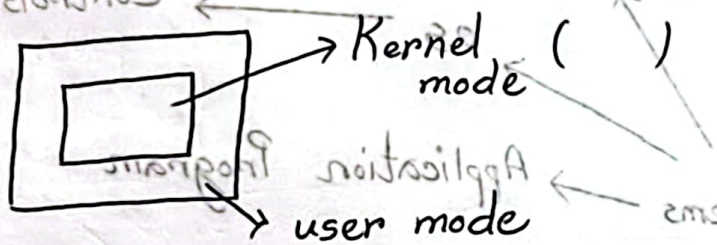
Introduction



* The hardwares that connects with computer connects with computer through a driver.

Operating System Defination

- User View → Uses its resources
  - Performance & Security
  - Program execution, communicatio
    error rate
- System View → Resource Allocator
  (CPU time, Memory, Space
  - Control Program
  - Protection & Security.



OS has mainly 2 parts.

(i) One that runs all the time in the system, is called Kernel.

(ii) Along with Kernel, there are 2 other types of prog. (1) Application Prog (All progra
not associat
with OS)

(ii) System Prog (Associated with
but are not necess

(Mainly part of Kernel)

- OS কী? Components
- Boot process কীভাবে হয়?
- DMA কীভাবে কাজ করে?

Booting Process

① When button is pressed, it supplies power on mother board

② This supply activates the booting process.

③ It then does Power of self test, that is stored in ROM

④ It checks if all hardware is connected properly

⑤ If everything OK, the system starts.

Memory

Primary Memory: Random access. typically volatile

Secondary Memory: Provides large non-volatile storage capacity.

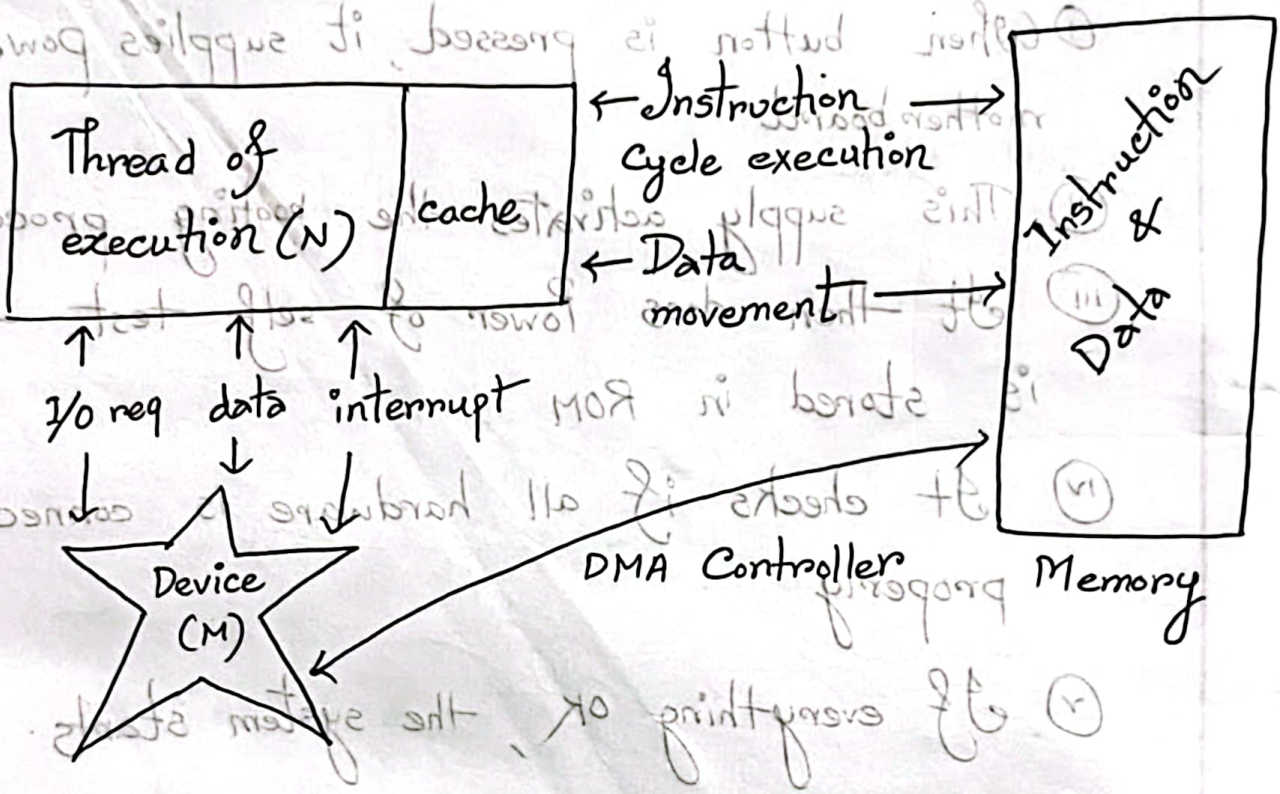
HDD → Hard Disk Drive

SSD

↓ Faster.

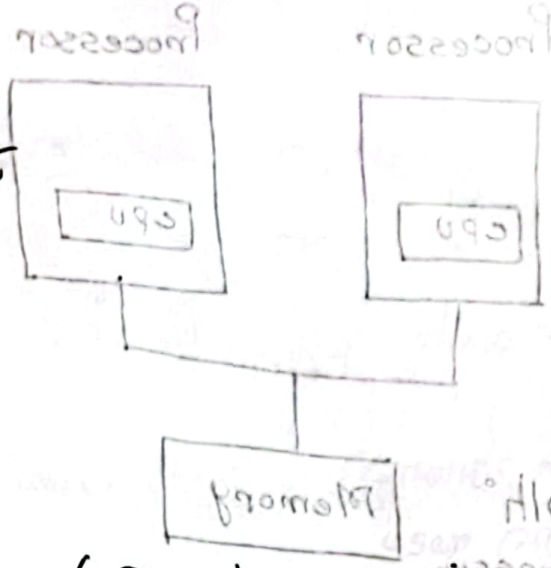
DMA

Direct Memory Access (DMA)



- Normally, execution goes through CPU
- But those execution that dont need execution or only help from CPU, in order to handle them, DMA is introduced.
- DMA directly connects I/O to memory.
- Throughput increase → একক সময়ে কাজের পরিমাণ

- OS Architecture
- OS Structure
- Process Management
- Memory "
- File "

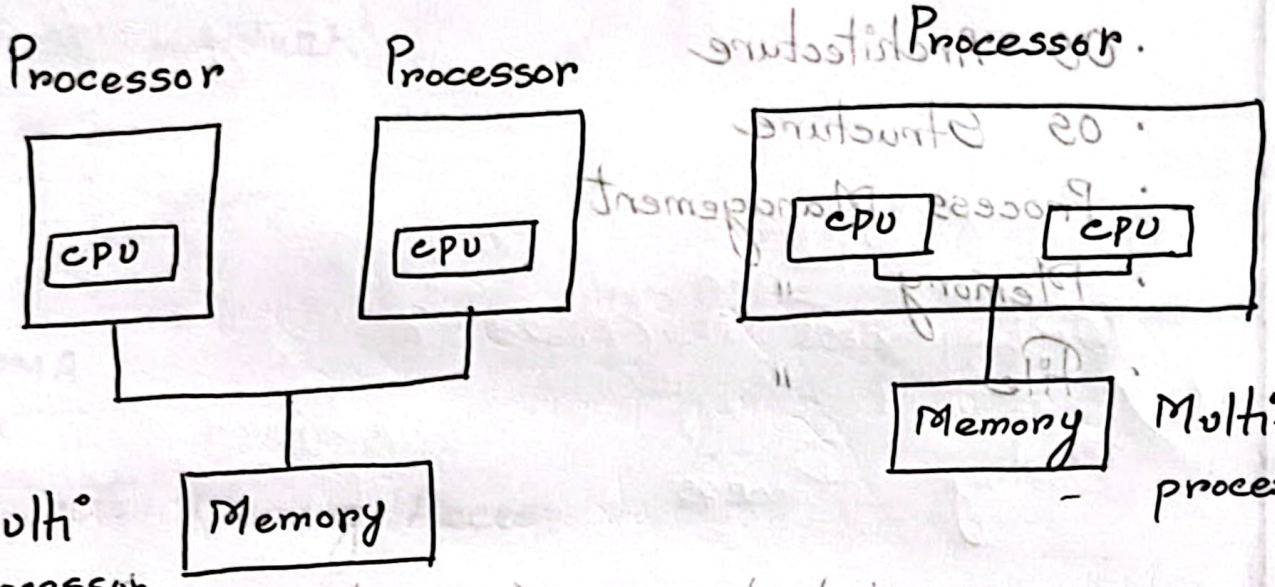


OS Architecture (Based on assignment)

- Symmetric → OS same কাজ সবারকে করতে হয়
- Asymmetric → Memory Slave / Different Process do different work
- Clustering → Some processes are assigned to specific task.

(Based on processor)

- Single Processor : Only 1 processor in the sys
- Multi Processor : Many single processor.
- Multi Core processor: More than 1 core within a single processor.



Multi-Cor
processor

(Based on assignment)

OS Architecture

OS ← ও → সব Multicore ই Multiprocessor, but সব Multiprocessor, Multicore নাও হতে পারে

• Clustering → Some processes are assigned specific task.

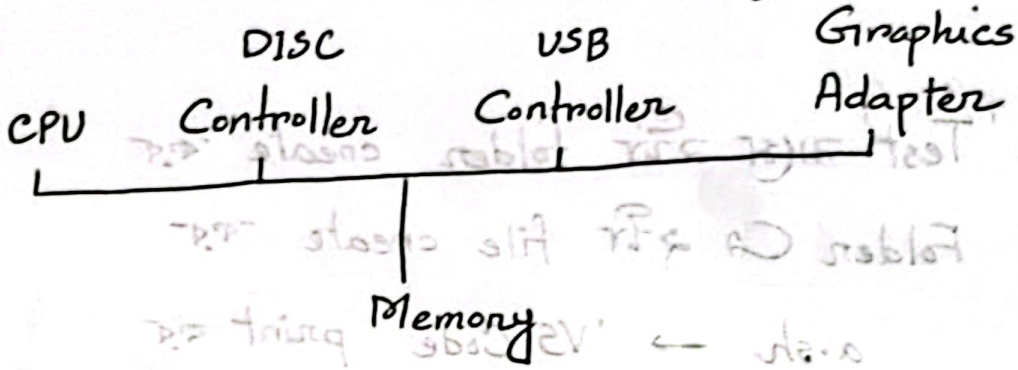
(Based on processor)

• Single Processor : Only 1 processor in the

• Multi Processor : Many single processor

• Multi Core processor : More than 1 core a single processor

Computer System Organization



With the help of DMA, I/O & CPU can execute concurrently. Each device controller type has an OS device driver to manage it.

Device Controller informs CPU that it has finished its operation by causing interrupt

(একটা কাজ হতে হতে আরেকটা কাজ আসা.)

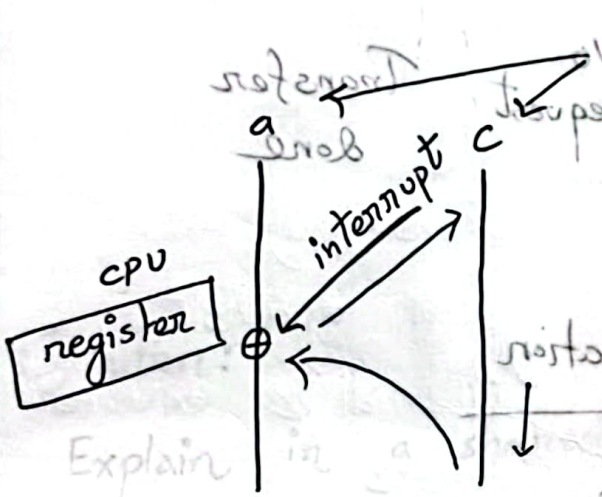
***

Interrupt

When interrupt occurs, CPU reads the interrupt num & jumps to interrupt vector, saving present coork in register

(II) Performs necessary processing. Performs a state transition & executes a return from statement.

(III) When returning, it temporarily pauses CPU's user process & performs system restore prior to interrupt.

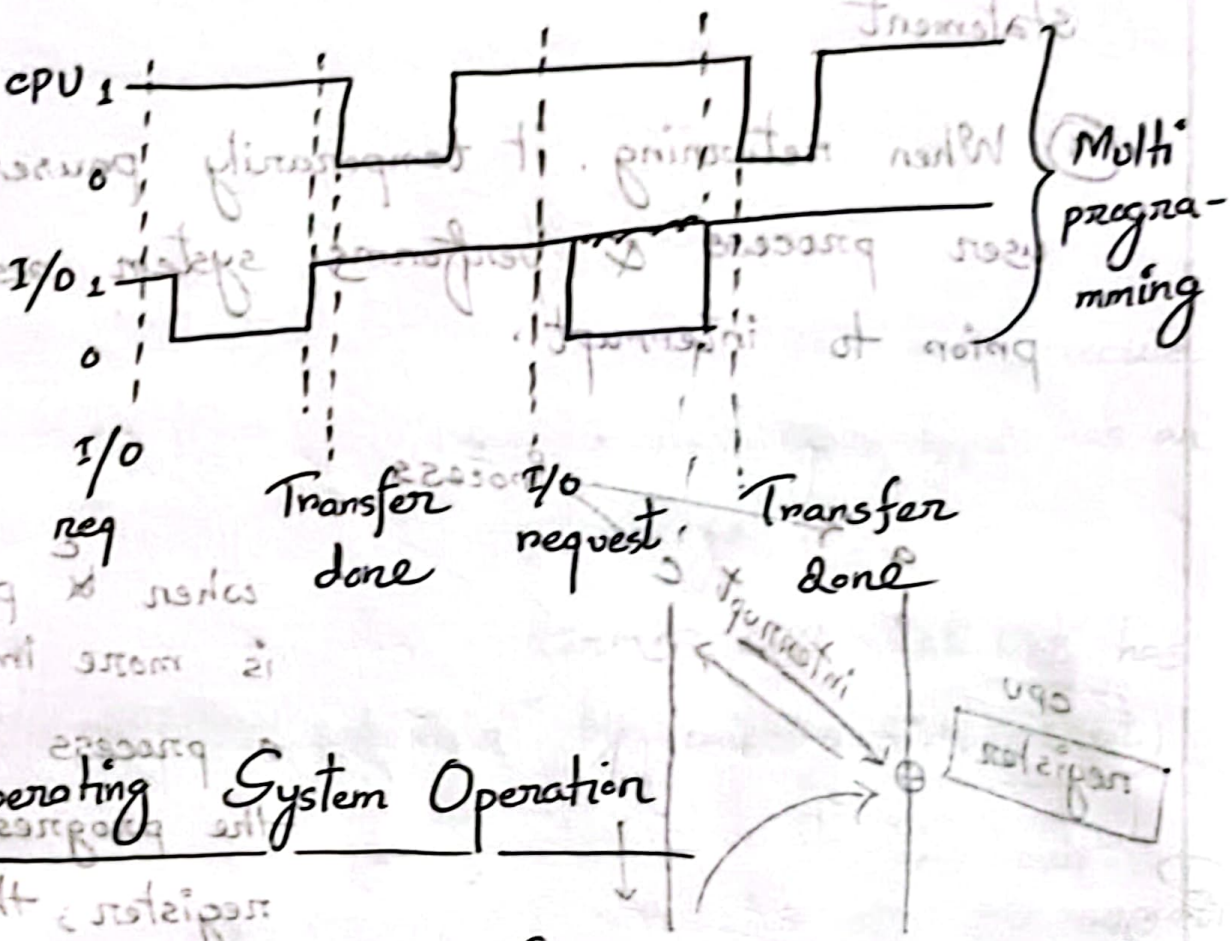


when a process is more important CPU stops & saves the progress in register; then c process starts. After c process is done, a process again starts.

- Interrupt Driven I/O Cycle
- Interrupt timeline

Slide → watch the diagrams.

Interrupt Timeline Diagram



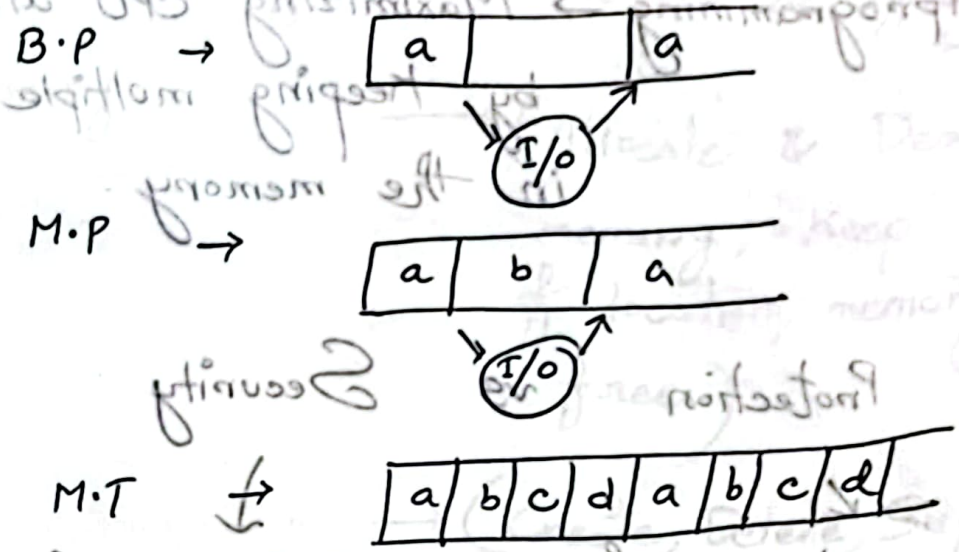
Operating System Operation

(i) Batch Processing (only 1 process can work)

(ii) Multiprogramming (Non-premptive, at a time more than 1 process, when once busy, others working)

(iii) Multitasking — Premptive time sharing
If several jobs ready to run at the same time

CPU does scheduling



Question:
Explain in a shared processing System, multitasking is more suitable than multiprogramming — (5)

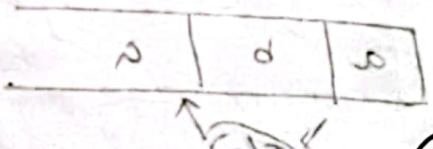
Non-Premptive → সামলে পারবে না
Premptive → মেমে মেমে কাজ করবে

Ans:- Allows multiple users to interact with the system simultaneously

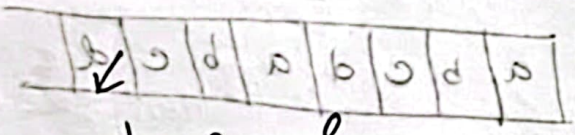
Multitasking → CPU time sharing, each user

gets a time slice more user interaction at a time, more responsive.

Multiprogramming → Maximizing CPU utilization by keeping multiple program in the memory.



Q→ Protection vs Security



Any mechanism for controlling access of processes or users to resources defined by

Defense of the system against internal & external attacks.

Ans:- Allows multiple users to interact with the system simultaneously

Resource Management

✓ Process Management → Create & delete files, control user & system process, thread & synchronization (বেকারিক case handle)

✓ Memory " → (Allocate & Deallocate memory, Keep track of location, memory use or free)

✓ file " → (Create, Delete, Support, Mapping, Backup)

✓ Storage " → (Allocation, Scheduling which process when run, partitioning protection)

✓ Cache " → (Faster Storage System, only most used are preserved)

✓ I/O " → (I/O related work)

☆ Computing → Just Reading

Q→ OS me How OS manages memory and file? (5 marks)

CT (Chap 1 & 3)

Date :- 22/1/26

Time :- 3:00 pm

Ch-03

An active program is a process. A process is the unit of work in a modern computing system.

Some attributes of process concept:

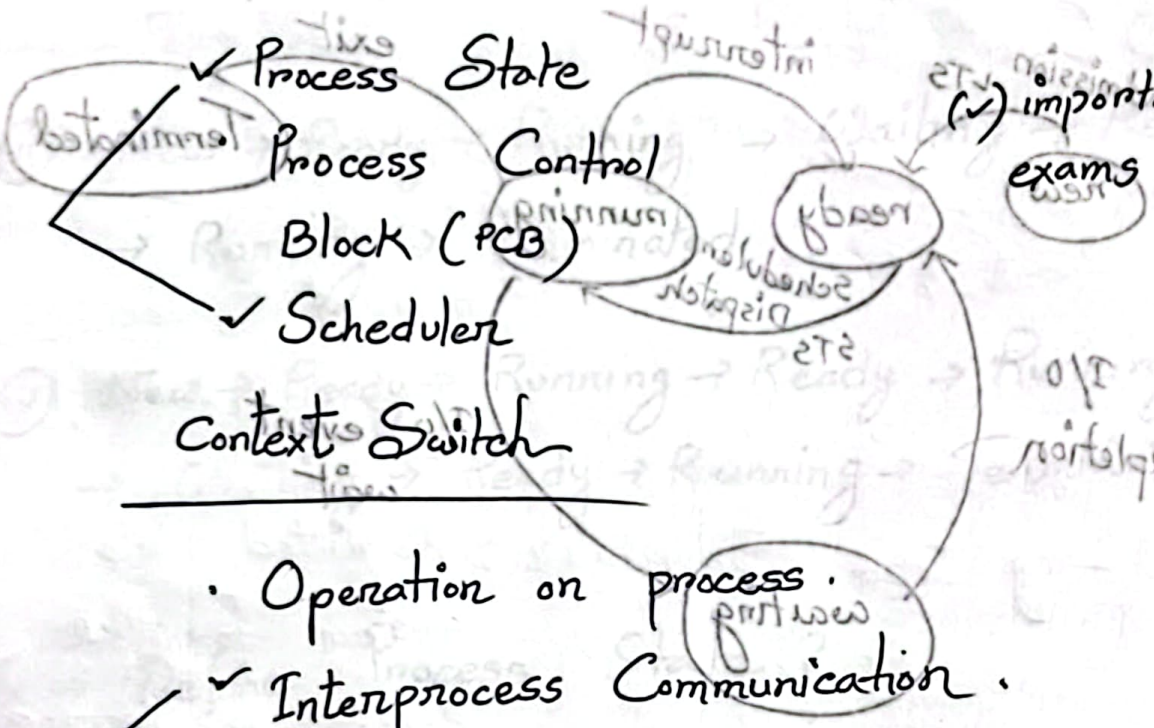
① Process ID (Unique name)
② Process State
③ Process Control Block
④ Threads

· A program becomes a process when an executable file is loaded into memory.

CH:03 → Topic

- ✓ Process State
- Process Control Block (PCB)
- ✓ Scheduler

(✓) important for exams



Context Switch

· Operation on process.

- ✓ Interprocess Communication.
- ✓ Remote Procedure Call.

Process State

---

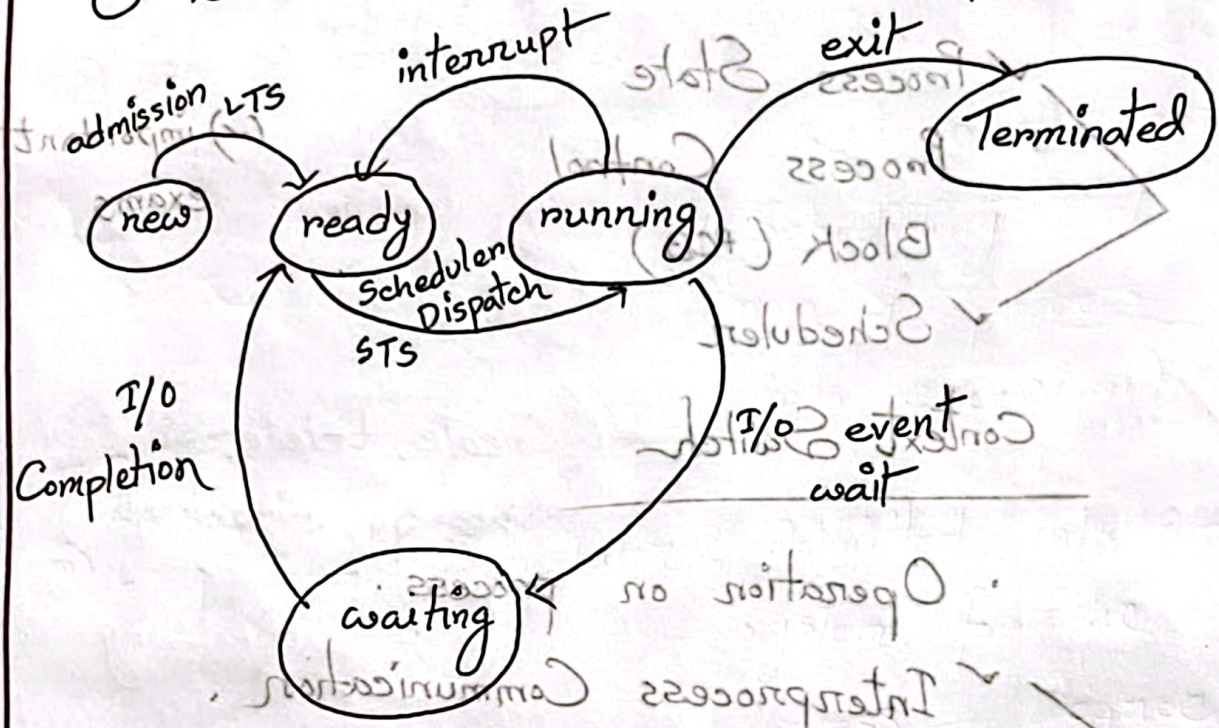
① New – Create new files in the memory / Existing files.

② Running – CPU তে run হয় যে process গুলো.

③ Ready – যে State গুলো CPU তে run হতে হয়, RAM এ load হয়.

(iv) Waiting - এর process মূলত CPU বাদে / others I/O related কাজে wait করে.

(v) Terminated - Process Run Complete



Scheduler is something which selects a process among various process.

Dispatcher is a module that gives control of CPU to the process selected by scheduler.

Possible Steps

(i) New – Ready – Running – Terminated

(ii) New – Ready – Running → Ready → Running → Terminated

(iii) New → Ready → Running → Waiting → Ready → Running → Terminated

(iv) New → Ready → Running → Ready → Running → Waiting → Ready → Running → Terminated

Q → Define Process State ?

Scheduler is something which selects process among various process.

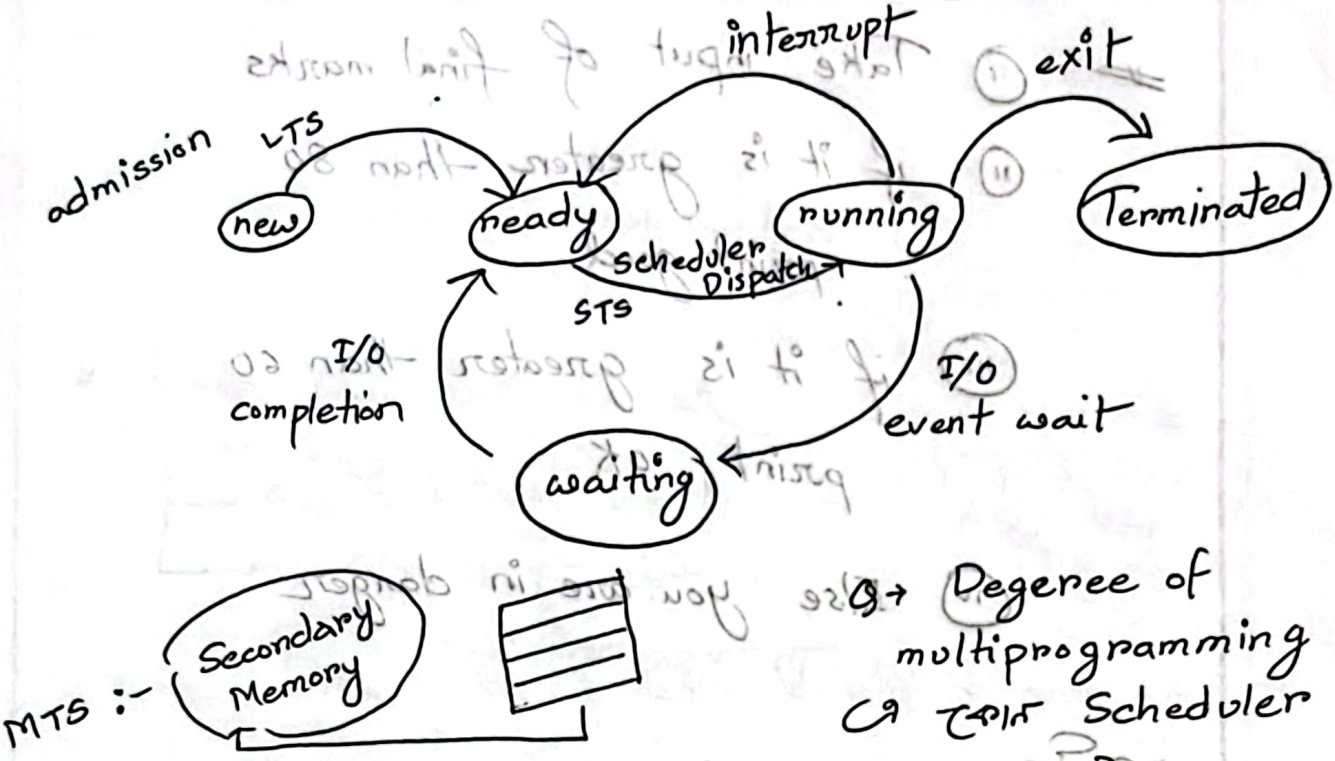
It can be of 3 types.

(i) Short term Scheduler: What will execute next on the CPU (Scheduling ctrl the num of process into execution)

Q → কীভাবে? কোনটা?

(ii) Long term Scheduler: Admitted into system for the Process → Degree of multipro-gramming

(iii) Medium Term Scheduler: Temporarily removes process from memory to reduce load & improve efficiency.



Q→ Degeree of multiprogramming কে কন্ট্রোল Scheduler কিভাবে করে?

Process Control Block

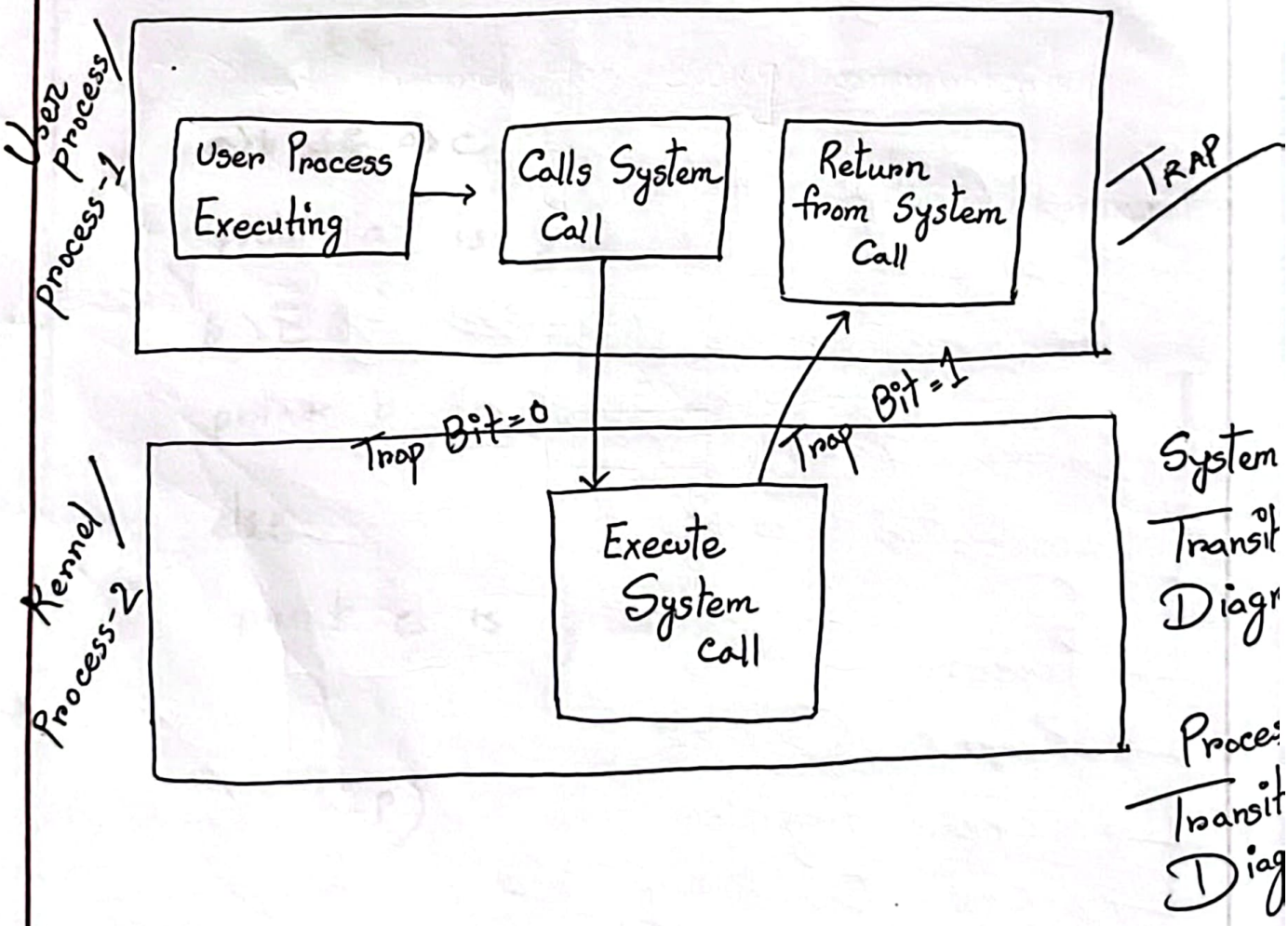
Each Process is presented by a Process Control Block (PCB). Each block is associated with

- Process State
- Process Counter
- CPU Register
- CPU Scheduling Info
- I/O Status

- Context Switching
- Interprocess Communication (IPC)
- Remote Procedure Call (RPC)
- Process Operation

Context Switching → $P_1 \rightarrow P_2$ → data transfer.

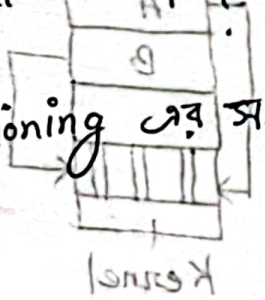
It is also called switching



- Process switch interrupt, I/O request or the process reaching its time quantum limit
- OS decides which process to run next
- All essential of running process saved. saves PCB of running process & loads the PCB of new process
- The CPU serves the content of new process & execute it
- Once complete it returns back to previous process.

Q → PCB (Process Ctrl Block) এর ভূমিকা কী ? (5 marks)

↑ Process Transitioning এর সময়



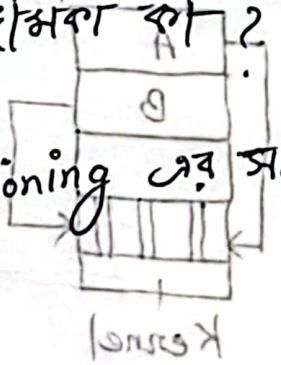
- Process switch interrupt, I/O request or the process reaching its time quantum limit.
- OS decides which process to run next.
- All essential of running process saved. saves PCB of running process & loads the PCB of new process
- The CPU serves the content of new-process & execute it.
- Once complete it returns back to previous process.

(Process Ctrl Block)

Q → PCB এর ভূমিকা কী ? (5 marks)

↑

Process Transitioning এর সময়

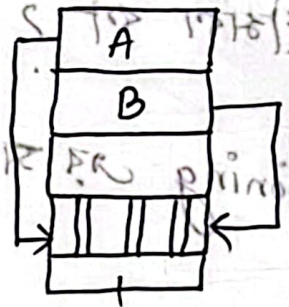


Inter Process Communication

একাধিক Process এর মধ্যে মাধ্যমে তথ্য/data Transfer হয়,

② Message Passing

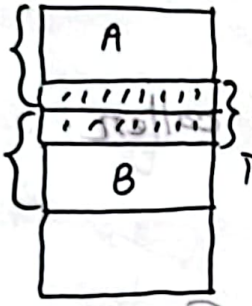
- Easier, Smaller data transfer. Only small amount of info transfer possible.
- A communication link must exist between (A & B) 2 process
- Need Kernel for every communication. Slow & costly.



System → Independent process
→ Concurrent process

③ Shared Memory

- Share a memory space directly.
- To share the memory, system call needs only 1 time.
- Easy but the processes must be in the same system.
- Faster



Remote Procedure Call (RPC)

P_1 → Server → data transfer

RPC is a message based communication schema to provide remote service.

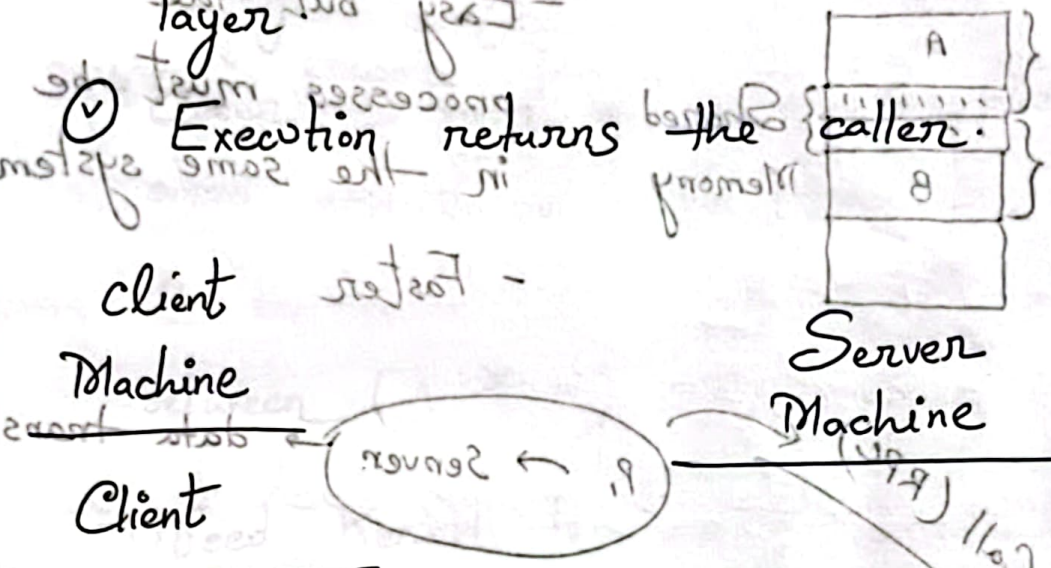
① When RPC is made, calling arguments are packaged & transmit from system to network server.

② The server unpacks the arguments, perform the desired procedure & sends the result back to clients.

(III) On server, transport layer pass the info to server stub

(IV) It hands over to transport layer, which give back to client transfer layer

(V) Execution returns the caller.



How Connection Established with Server?

(i) Sender System Needs to communicate with a process on receiver. However, sender doesn't know the specific port number. The issue is resolved with RPC runtime system, which handles communication & port assignment dynamically.

(ii) When a server start, it registers its service with RPC binding Registry along with a port number. This allows the server to be accesible by client.

(iii) Client Contacts the RPC runtime system to get the port number. Once binding is done they can communicate through that port.

Q → (i) How RPC works?

(ii) How connection established between client & server system?

CT markings → 3x5 que → 15 marks

Process Operation:

① Creation:

fork() → Creates a new process

exec() → Runs an executable file

wait() → Suspends execution of current process until 1 of the children terminates.

② Terminate

→ When process ends after completion

→ Force the system to stop

Parent → fork()

Parent (PID=1) → wait()

Parent resume

↳ Child → exec() → exit()

(=PID) (PID)

new, running
ready, wait
terminated ← Process State / Scheduler / CPU Scheduling → LTS, STS, MTS

Scheduler (LTS)

ch-03

- Explain attributes of process.
- Explain diff types of Scheduler: Where & for which task are they used
- When we use RPC? How it works? How connection happens.
- Context Switching?
- Interprocess Types. Where, which better?

ch-01

- Explain how OS manages file / Memory / Process.
- Define OS based on user perspective & system view.

T = C - A

- How booting process work.
- How interrupt happens with diagram. Describe its timeline diagram.

W = T - B

- How DMA is used; Compare multiprocessor vs

Priority Scheduling

SRTF is a special case of priority Scheduling where shortest time is main priority.
Can be either pre-emptive or non-premptive
A priority scheduling algo will pre-empt the CPU if priority of newly arrived process is higher than the priority of currently running process.

☆ A major prob with PR is infinite blocking or starvation. Higher priority process can prevent a lesser lower priority process that wait in the Sys for a long time.

FCFS → Convoy effect
  ↳ Simple

SRTF → Calculation বেশি
  ↳ Avg waiting time কম

Soln: Gradually increament the priority after a fixed time

Round Robin

Time Sharing Algo

Use a small unit of time - called Quantum/Slice time

Performance of RR highly depends on the value of quantum.

* Quantum অনেক বড় হলে FCFS এর মতো কাজ করে।
* Quantum too small হলে,
  (i) Context Switch Process ↑
  (ii) Response time decreases.
  (iii) Change of Starvation decrease

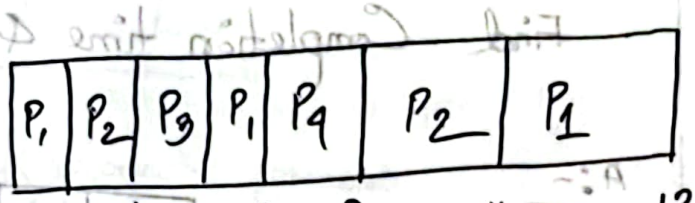
	A	B
P_1	0	~~5~~ ~~2~~ ~~1~~
P_2	1	~~4~~ ~~3~~ 0 1
P_3	2	~~2~~ 0
P_4	4	~~3~~ 0

★ নতুন কী কী process আসছে

★ Running আবার লাগলে লিখব ই।

$Q = 2$

Gantt Chart

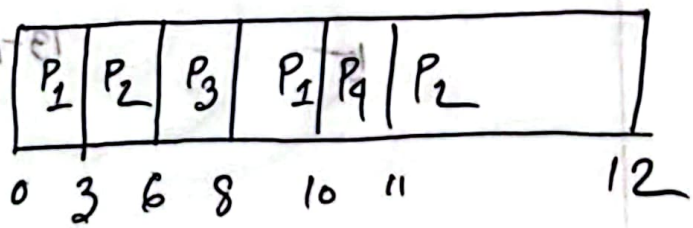


Com	Res
12	0-A
11	2-A
6	4-A
9	8-A

Rt
A-0
A-3
A-2
A-1

Q
3
1
1
1

$Q = 3$



A	B
0	10
1	1
1	10
4	1
9	20

Q=3

Find Completion time & Response time ?

A:-

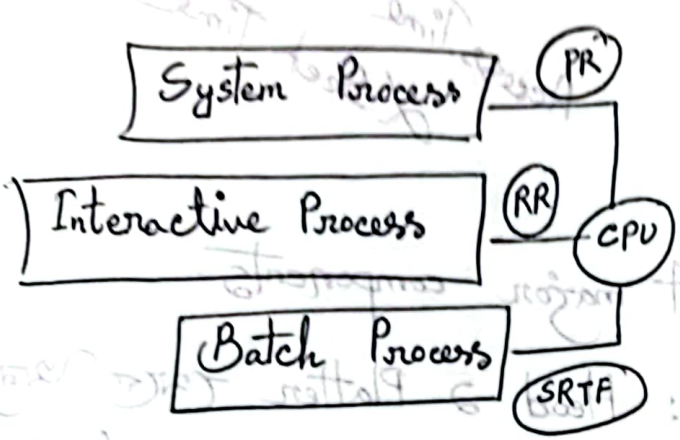
Gantt Chart

P1	P2	P3	P4	P2	P5	P3	P2	

0 3 6 9 10 13 15 18 19

Cm	Rt
3	0-A
19	3-A
18	6-A
10	9-A
15	13-A

Rt	Cm
A-0	15
A-3	11
A-6	2
A-9	5

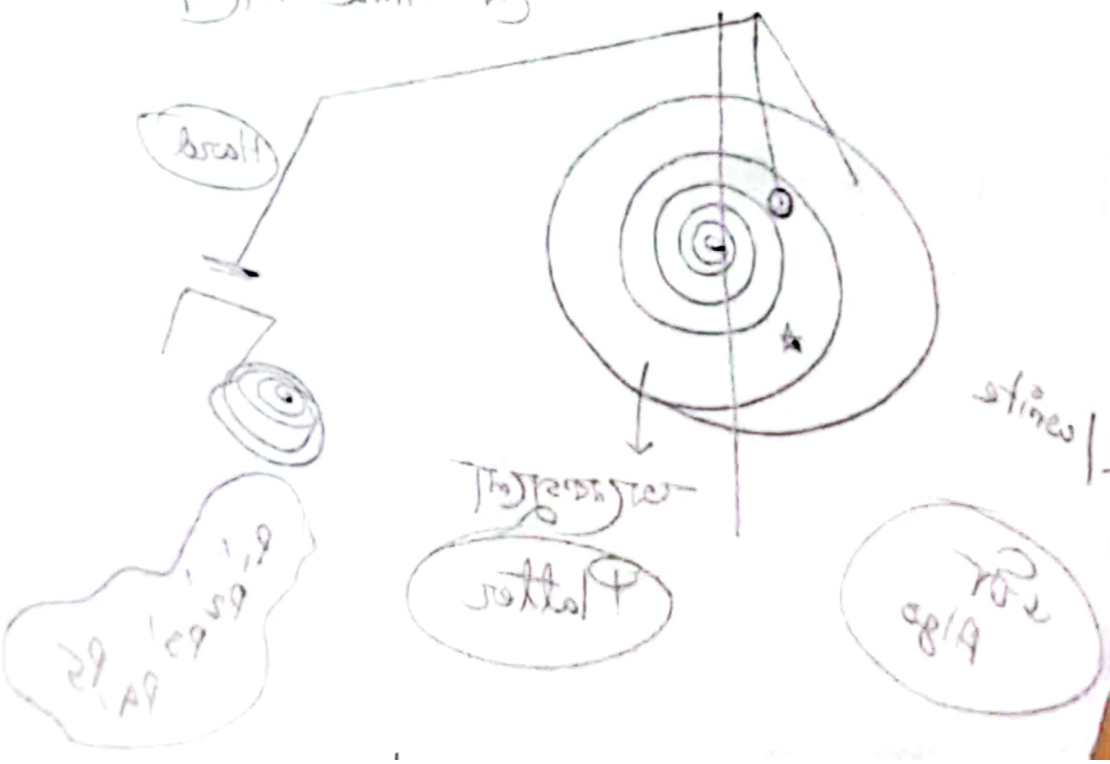


(v) Multilevel Queue Scheduling

Foreground & Background process

(vi) Multilevel feedback queue Scheduling:

Allows a process to move between queues.



Disc Scheduling

Access Time = Seek Time +

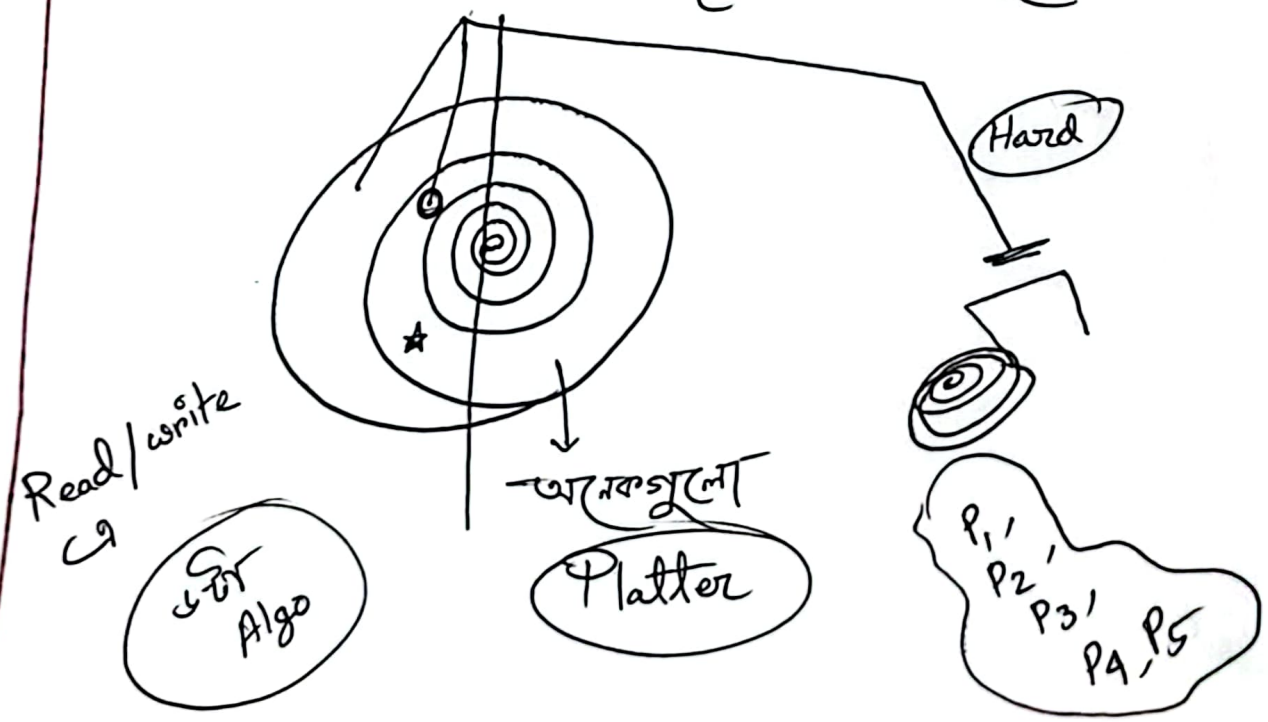
Access time has 4 major components.

(i) Seek time : Head ২ Platter থেকে অন্য Platter এ যেতে যে time.

(ii) Rotational Latency : Desired Sector যেখানে Read / Write করবে।

(iii) Head Activation Time : Preparation to start read / write.

(iv) Transfer Time : যেখানে Read / write করতে যে time লাগে।



Algo

(i) FCFS

(ii) SSTF — Shortest Seek time first

(iii) SCAN

(iv) C-SCAN

(v) Look

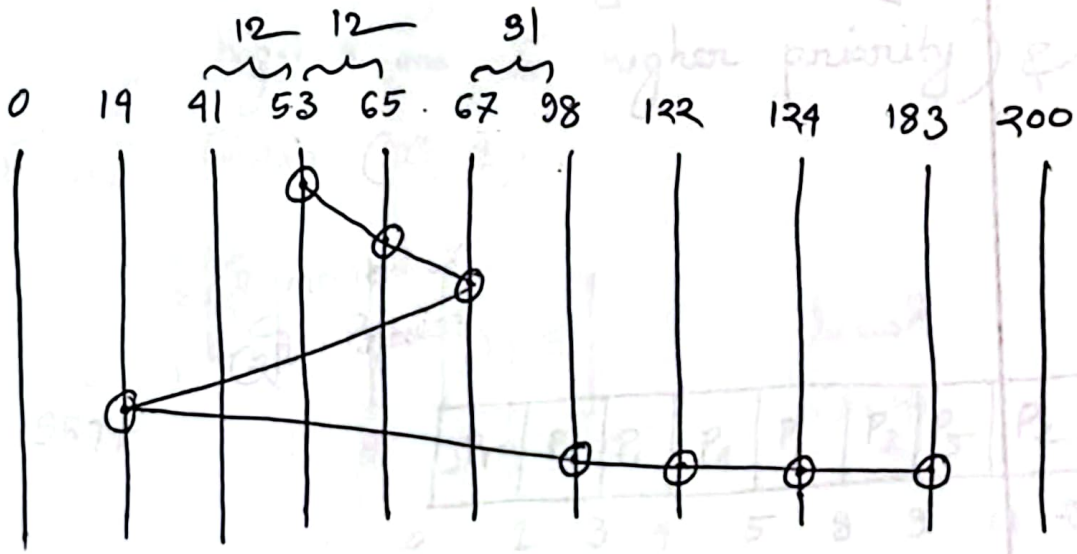
(vi) C-Look

Req → 98, 183, 41, 122, 14, 124, 65, 67

Platter (0-200)

Start - 53

Seek / Total Acce
Time Comp



বাকি সব Same

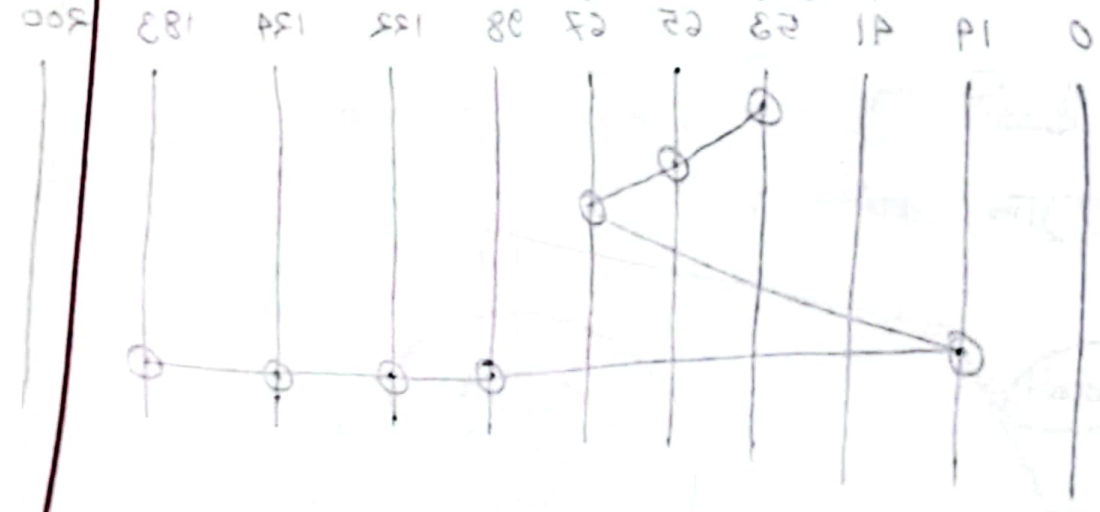
$|65-41| = 24$

$|14-41| = 37$

$|65-67| = 2$

$|65-41| = 24$

Total Seek Time $= (67-53)+(67-14)+(183-14)$

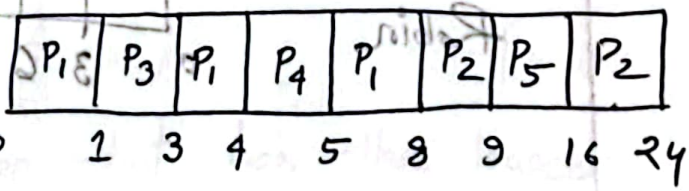


Arrival Time	Burst Time	Priority
0	~~5~~ 4	4
1	9	1
1	2	2
4	1	5
9	7	3

① Perform avg turnaround time, avg waiting time, & avg response time using Shortest Running Time First (SRTF), Priority Scheduling (Taking highest one as higher priority) & Round Robin (Q=3)

A:- → moment of Process

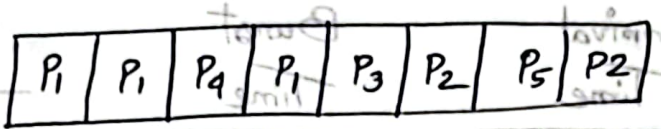
SRTF



Com	Res
8	0-0=0
24	8-1=7
3	1-1=0
5	4-4=0
16	9-9=0

avg Response time = 1.4

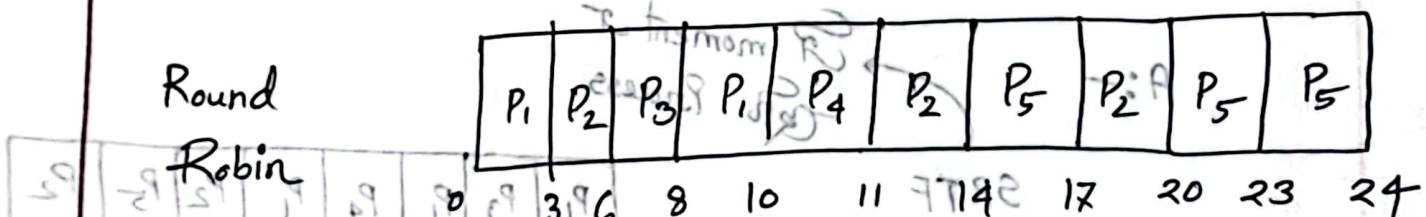
Priority Scheduling



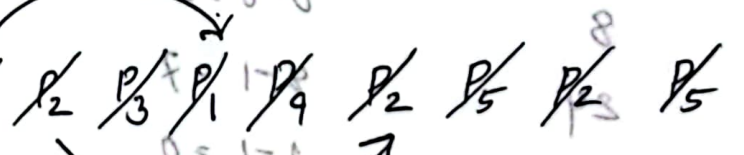
Com	Res	T.C-A	W = T-B
6	0-0=0	6-0=6	6-5=1
24	8-1=7	24-1=23	23-9=14
8	6-1=5	8-1=7	7-2=5
5	4-4=0	5-4=1	1-1=0
16	9-9=0	16-9=7	7-7=0

avg W = 4

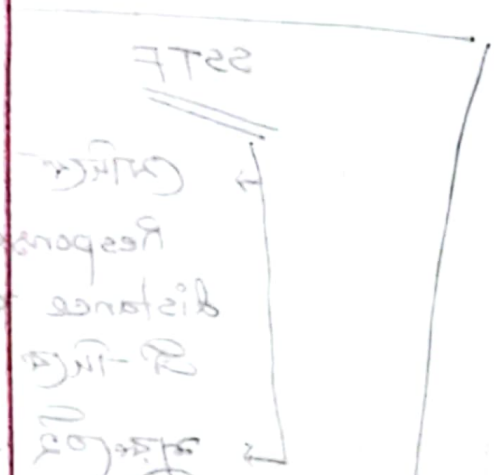
Round Robin (Taking Q = 3)



Ready Queue



Com	Res	T=C-A	W=T-B



(11) Which algo is better in this case justify your ans?

A:- → Avg TA
→ Avg WT
→ Avg Res
} for SRTF, PR, RR

→ The algo that has the lowest WT is better.

Disc Scheduling

Req : 52, 93, 100, 140, 20, 32, 86, 93, 71

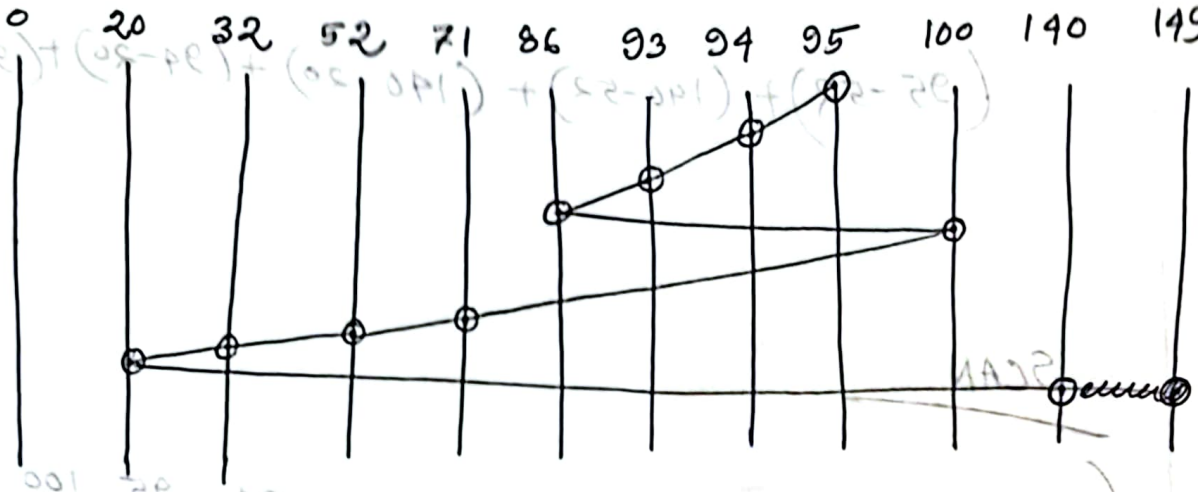
Platter : 0 – 149

Start : 95

SSTF

- সেদিকে Response distance কম সে-দিকে যাব.
- শুরুতেই ২ দিকে same distance হলে, তারপর কোন Req এ distance কম, সেটা নিব, পূর্বের গুলোতে একদিকে নিলেই হবে.

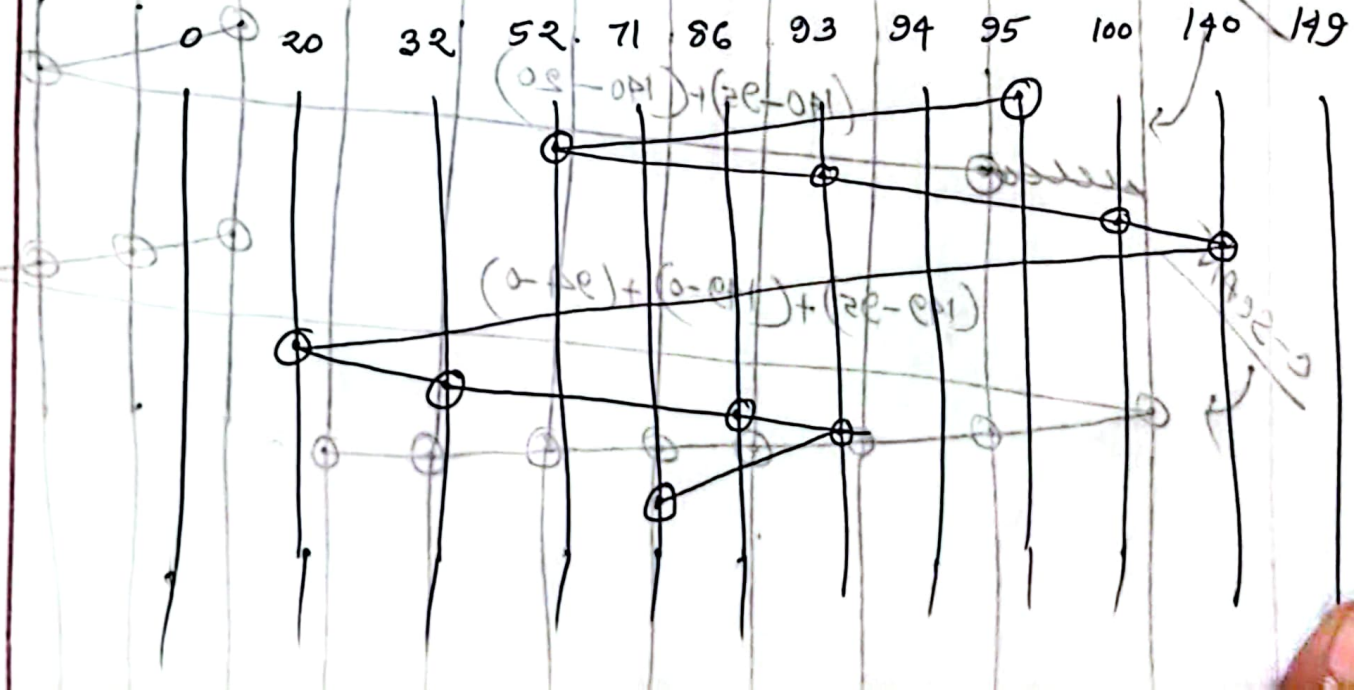
① Sorting



Total Seek Time

$= (95-86) + (100-86) + (100-20) + (140-20)$

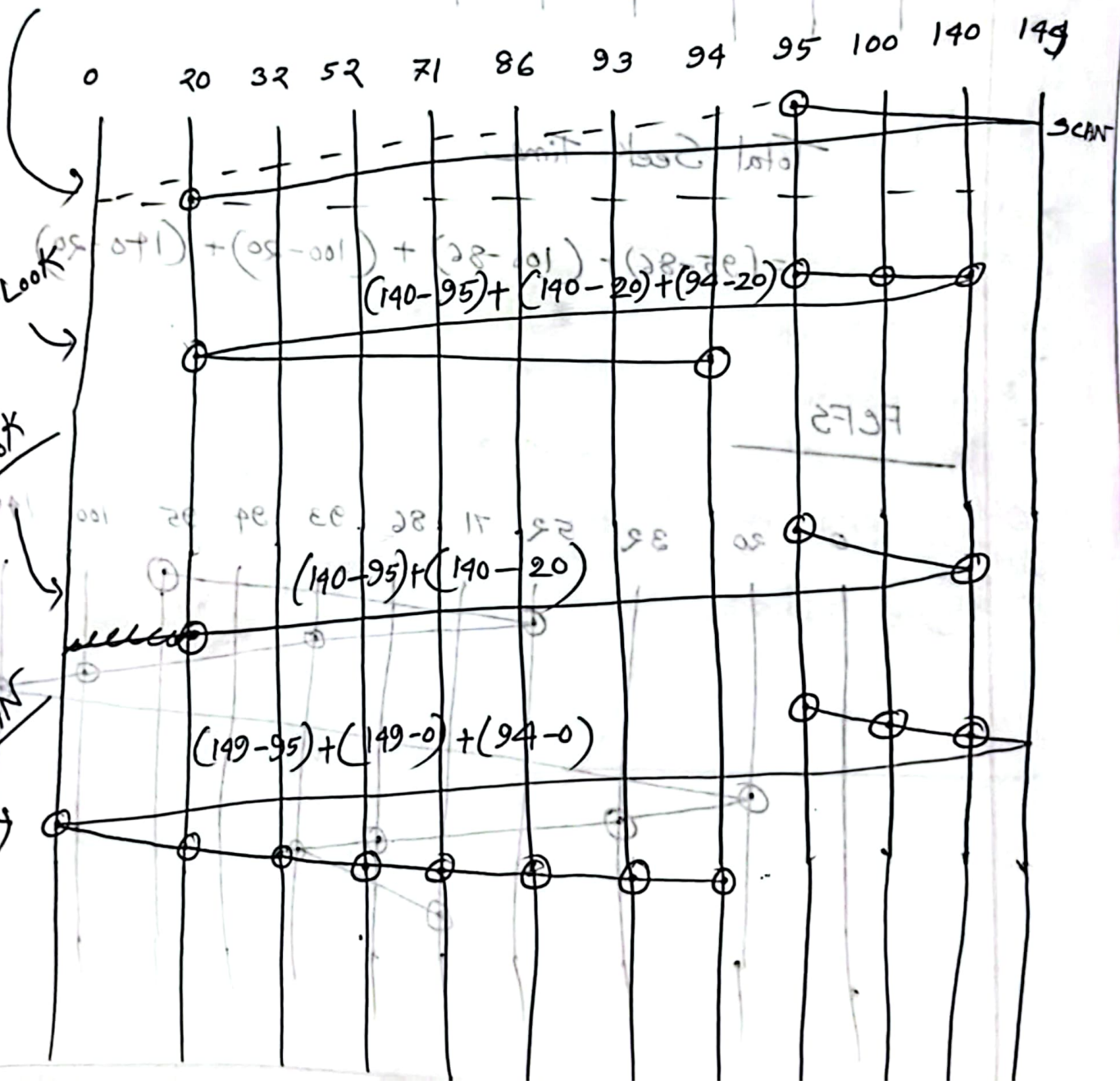
FCFS



Total Seek Time

$(95-52)+(140-52)+(140-20)+(94-20)+(94-0)$

SCAN



C-LOOK

$(140-95)+(140-20)+(94-20)$

LOOK

$(140-95)+(140-20)$

C-SCAN

$(149-95)+(149-0)+(94-0)$

First ~~Come~~ First Serve

Look - Scan এর update version
- Platter touch না করেই Back করে।

C-Look - C-Scan এর ..... by default ভালো আগে

☆ SCAN, C-SCAN → Platter touch করে, বাকিরা করে না।